

Indian Institute of Technology (ISM), Dhanbad

Department of Computer Science and Engineering
Monsoon Mid Semester Examination, 2025-26

Examination: BTech Ist Semester
Subject: Computer Programming (NCSV101)

Max. Marks: 30
Time: 1 Hour

Instructions:

- Answer all the questions.
- Use two separate answer booklets, one for each part.
- Write the part number at the top of the front page of each answer booklet.

I have read the instructions mentioned above in this question paper.

Part I (15 marks)

1. Write a C program that accepts a number from the user (between 1 and 7). Each number corresponds to a day of the week, where 1 represents Sunday and 7 represents Saturday. The program should display the name of the day according to the input. If the user enters any number outside the range 1–7, the program should print "Invalid day". (4)
2. Write a C program that accepts an integer k from the user. The program should print a **right-angled triangle** pattern of stars (*) with k rows. (3.5)

k=2	k=3	k=4
*	*	*
**	**	**
	***	***

3. What will be the output of the following code? Explain the output briefly. (3)

```
#include <stdio.h>
int main() {
    int a, b=5;
    int x = (a=0) && (b=4);
    printf("%d %d %d", x, a, b);
}
```

4. Write the decimal number 260 in binary, octal, and hexadecimal representations. Show the calculations. (4.5)

Part II (15 marks)

5. Predict the output of the following C program: (2.5)

```
#include <stdio.h>
int main() {
    int a = 5, b = 2; float x;
    x = (a / b) + (float)(a % b) / b;
    if (x > 2.0)
        printf("Result = %.2f\n", x);
    else
        printf("Value too small\n");
}
```

6. Write a C program that reads an integer n and prints all numbers from 1 to n in reverse order using a *do-while* loop. If the user enters a value of n as zero or a negative value, the program must print "Invalid input" and exit. **NOTE: Marks will be awarded only if the program successfully returns the required output on a computer. (5)**
7. Draw a flow chart to read an integer from the user and find the sum of the digits of that integer. (3)
8. Write a C program to read 20 integers from the user and store those integers in an array. Print the difference between the sum of odd integers and the sum of even integers in the array. Print the average of odd integers in the array. (4.5)

e

$$e^{\frac{\ln x}{x}}$$

$$\frac{1}{x} - x$$

$$0! = 1$$